

**PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)
ON**

**Comparative Analysis of Machine Learning and Deep Learning Models for
Sentiment Analysis**

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**Comparative Analysis of Machine Learning and Deep Learning Models for Sentiment Analysis**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the Department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Shikha Tuteja**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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ABSTRACT

The expansion of online textual data has made sentiment analysis a crucial task in Natural Language Processing (NLP). This study presents a comprehensive comparative analysis of traditional Machine Learning (ML) models and Deep Learning (DL) models for sentiment classification on the IMDb movie reviews dataset. The ML models examined include Naive Bayes, Logistic Regression, and Support Vector Machine (SVM), while the DL component is represented by a Long Short-Term Memory (LSTM) network; transformer-based BERT is also discussed as a benchmark for future direction.

Models were evaluated using standard metrics, namely Accuracy, Precision, Recall, and F1-score, alongside computational efficiency in terms of training time and resource usage. Experimental results show that LSTM achieves the highest overall performance with an accuracy of 90 percent, outperforming all ML baselines. Among the traditional models, SVM ranks first with 86 percent accuracy, followed by Logistic Regression at 85 percent and Naive Bayes at 82 percent.

The findings demonstrate a clear trade-off between accuracy and computational cost: deep learning models deliver superior contextual understanding but require substantially more training time and resources, whereas machine learning models remain effective and practical in resource-constrained or real-time scenarios. The report concludes by recommending hybrid and transformer-based approaches such as BERT and GPT, along with optimisation techniques including pruning, quantisation, and knowledge distillation, as promising directions for future work.

Keywords: *sentiment analysis, machine learning, deep learning, LSTM, BERT, NLP, SVM, Naive Bayes, logistic regression, text classification.*

1. INTRODUCTION

1.1 Background

Sentiment analysis, also known as opinion mining, has emerged as a crucial task in the field of Natural Language Processing (NLP), driven by the rapid growth of user-generated content on digital platforms such as social media, e-commerce websites, and online forums. With the proliferation of textual data, organisations and researchers are increasingly interested in automatically identifying and extracting subjective information, including opinions, attitudes, and emotions expressed in text. Applications of sentiment analysis span diverse domains, including customer feedback analysis, brand monitoring, political forecasting, and decision support systems, making it a significant area of study within both academia and industry.

1.2 Traditional Machine Learning Approaches

Traditionally, sentiment analysis has relied on Machine Learning (ML) techniques such as Naive Bayes, Support Vector Machines (SVM), and Logistic Regression. These approaches typically involve a structured pipeline consisting of data preprocessing, feature extraction, and model training. Feature engineering plays a central role in ML-based models, where textual data is transformed into numerical representations using methods like bag-of-words, n-grams, and term frequency – inverse document frequency (TF-IDF). While these models are computationally efficient and perform well on smaller datasets, their effectiveness often depends heavily on the quality of handcrafted features and domain-specific knowledge.

1.3 Emergence of Deep Learning

In recent years, Deep Learning (DL) models have revolutionised sentiment analysis by enabling automatic feature learning from raw text data. Architectures such as Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, and Convolutional Neural Networks (CNNs) have demonstrated significant improvements over traditional methods, particularly in capturing contextual and sequential information in text. Furthermore, the advent of transformer-based models like BERT (Bidirectional Encoder Representations from Transformers) and GPT has further enhanced the performance of sentiment analysis systems by leveraging attention mechanisms and large-scale pretraining on vast corpora.

1.4 Motivation and Scope

Despite their superior performance, deep learning models come with certain limitations, including high computational requirements, longer training times, and the need for large annotated datasets. In contrast, traditional machine learning models remain relevant due to their simplicity, interpretability, and efficiency, especially in resource-constrained environments. Therefore, a comparative analysis of ML and DL approaches is essential to understand their respective strengths, limitations, and suitability for different sentiment analysis tasks.

This report aims to provide a comprehensive comparison between machine learning and deep learning models for sentiment analysis. It examines their performance across the IMDb dataset, evaluates key metrics such as accuracy, precision, recall, and F1-score, and discusses factors such as scalability, interpretability, and computational complexity.

1.5 Literature Review

Sentiment analysis has been a prominent research area within NLP, driven by the exponential growth of user-generated content across digital platforms. Over the years, researchers have explored a wide range of approaches, broadly categorised into lexicon-based methods, traditional ML techniques, and more recently, DL models.

Early research in sentiment analysis primarily relied on lexicon-based and rule-based approaches, which utilised predefined sentiment dictionaries to determine polarity. However, these methods often struggled with contextual understanding, sarcasm, and domain-specific language. As a result, ML techniques gained popularity due to their ability to learn patterns from data. Traditional ML models such as SVM, Naive Bayes, Logistic Regression, and Decision Trees became widely adopted for sentiment classification tasks.

With advancements in computational power and the availability of large datasets, deep learning models have increasingly dominated sentiment analysis research. Deep learning architectures such as CNNs, RNNs, and LSTM networks have demonstrated superior performance by automatically learning hierarchical representations of text data. Comparative studies indicate that transformer-based models, particularly BERT, consistently outperform traditional approaches due to their ability to capture bidirectional context and semantic nuances. Reviews of hybrid and ensemble

approaches further suggest that combining ML and DL techniques can improve performance while mitigating individual limitations.

1.6 Objectives

The objectives of this study are as follows:

- To perform a comparative analysis of traditional machine learning models (Naive Bayes, Support Vector Machines, and Logistic Regression) and deep learning models (CNN, LSTM, and transformer-based models) for sentiment analysis tasks.
- To evaluate and compare the performance of these models using standard evaluation metrics, including accuracy, precision, recall, and F1-score.
- To analyse the computational efficiency of machine learning and deep learning approaches in terms of training time, memory usage, and resource requirements.
- To identify the most suitable models for different scenarios, such as small versus large datasets, real-time applications, and resource-constrained environments.

2. METHODOLOGY

This study adopts an experimental methodology to compare the performance of machine learning and deep learning models for sentiment analysis. The process includes dataset preparation, preprocessing, feature extraction, model training, evaluation, and comparative analysis.

2.1 Dataset Selection

A standard sentiment analysis dataset is used, consisting of labelled text samples categorised as positive or negative. The IMDb movie reviews dataset was selected because of its size, balanced class distribution, and widespread use as a benchmark in sentiment analysis research. The dataset is divided into a training set comprising 70 percent of the samples and a testing set comprising the remaining 30 percent. Formally, the dataset can be represented as $D = \{(x_i, y_i)\}$, where x_i represents an input text and y_i represents the corresponding sentiment label.

2.2 Data Preprocessing

Before model training, the raw text data is cleaned and normalised through the following preprocessing steps: conversion to lowercase, removal of punctuation and special characters, stopword removal, tokenisation, and lemmatisation or stemming. These steps help reduce noise and improve model performance by standardising the input text and reducing vocabulary size.

2.3 Feature Extraction

For machine learning models, textual data is converted into numerical form using two complementary representations:

- Bag of Words (BoW), in which each document is represented by the frequency of words it contains, ignoring grammar and order.
- Term Frequency – Inverse Document Frequency (TF-IDF), which weights words by their frequency in a document and their rarity across the corpus, giving higher importance to discriminative terms.

For deep learning models, word embeddings are used to represent text in dense vector form, capturing semantic relationships between words and reducing the need for manual feature engineering.

2.4 Model Implementation

The following machine learning models are implemented: Naive Bayes, Support Vector Machine (SVM), and Logistic Regression. These models rely on extracted features and are computationally efficient. The deep learning models used in this study include the Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM), which automatically learn features from text and are effective in capturing contextual and sequential patterns. Each model is tuned using appropriate hyperparameters to improve performance:

- Naive Bayes — a probabilistic classifier based on Bayes’ theorem with feature independence assumptions.
- Logistic Regression — a linear model for binary classification using the sigmoid function.
- Support Vector Machine (SVM) — a margin-maximising classifier effective in high-dimensional spaces.
- Long Short-Term Memory (LSTM) — a recurrent neural network capable of learning long-range sequential dependencies.
- BERT — a transformer-based model leveraging bidirectional attention for deep contextual understanding (discussed as benchmark).

2.5 Evaluation Metrics

To compare the performance of models, standard evaluation metrics are used, including a Confusion Matrix and the following derived performance measures:

- $\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$
- $\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$
- $\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$
- $\text{F1-Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$

Table 1. Confusion Matrix Structure

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

3. TOOLS AND TECHNOLOGIES

The successful implementation of this comparative study required a combination of programming languages, libraries, frameworks, and development environments. This chapter outlines the tools and technologies used during the project.

3.1 Programming Language

Python was chosen as the primary programming language because of its rich ecosystem of libraries for natural language processing, data analysis, machine learning, and deep learning. Its readable syntax and active community made it well suited to rapid experimentation and reproducible research.

3.2 Machine Learning Libraries

- scikit-learn — used for implementing Naive Bayes, Logistic Regression, and Support Vector Machine classifiers, as well as utilities for train-test splitting, cross-validation, and metric computation.
- NumPy and Pandas — used for efficient numerical computation and tabular data manipulation.
- NLTK and spaCy — used for text preprocessing tasks, including tokenisation, stopword removal, lemmatisation, and stemming.

3.3 Deep Learning Libraries

- TensorFlow and Keras — used for building and training the LSTM model with high-level APIs that simplify layer composition, training, and evaluation.
- Hugging Face Transformers — referenced for transformer-based models such as BERT, providing pretrained weights and tokenisers for future extension of this work.

3.4 Visualisation and Analysis Tools

- Matplotlib and Seaborn — used to plot model performance, confusion matrices, and training curves.
- Jupyter Notebook — used as the interactive development environment for prototyping, running experiments, and presenting results alongside narrative explanations.

3.5 Development Environment and Hardware

Experiments were carried out on a workstation running a 64-bit operating system with a multi-core CPU, sufficient RAM for in-memory dataset processing, and a GPU available for accelerating LSTM training. For exploratory work, cloud-based environments such as Google Colab were also considered to provide free access to GPU resources.

3.6 Version Control and Documentation

Git was used for source code version control, with the project repository hosted on a remote platform to enable collaboration among team members. Markdown and LaTeX were used for documenting experiments, while this report itself was prepared as part of the deliverables for the IOHE programme.

4. IMPLEMENTATION

This chapter describes the practical implementation of the methodology described earlier. It covers the workflow from data acquisition through training and evaluation.

4.1 Data Acquisition and Preparation

The IMDb movie reviews dataset was acquired in its standard form, comprising labelled examples of positive and negative reviews. The data was loaded into a Pandas DataFrame for inspection. Class balance was verified, and a stratified split into training (70 percent) and testing (30 percent) subsets was performed to ensure that both classes were proportionally represented in each subset.

4.2 Preprocessing Pipeline

A preprocessing pipeline was constructed in Python that applied the following steps to every review in sequence: conversion to lowercase, removal of punctuation and non-alphanumeric characters, stopwords removal using a standard English stopwords list, tokenisation, and lemmatisation. The cleaned text was then ready for vectorisation. This pipeline was implemented as a reusable function so that the same transformations could be applied consistently to training and test data, as well as to any new text submitted for inference.

4.3 Feature Engineering for Machine Learning

For the ML models, two feature representations were generated from the preprocessed text. A Bag of Words representation was computed using a CountVectorizer, while a TF-IDF representation was computed using a TfidfVectorizer. Vocabulary size, n-gram range, and minimum document frequency were treated as hyperparameters and tuned to balance model accuracy with computational cost.

4.4 Training Machine Learning Models

Three traditional models were trained on the TF-IDF features:

- Naive Bayes was trained using the Multinomial variant, which is well suited to discrete text features.
- Logistic Regression was trained with L2 regularisation, with the regularisation strength tuned via cross-validation.

- SVM was trained using a linear kernel, which is known to perform well on high-dimensional sparse text features.

Each model was evaluated on the held-out test set using the metrics described in Chapter 2.

4.5 Building and Training the LSTM Model

The deep learning component centred on a Long Short-Term Memory (LSTM) network implemented in TensorFlow with the Keras API. The architecture consisted of an embedding layer that mapped tokenised reviews into dense vectors, a recurrent LSTM layer that processed the sequence and captured long-range dependencies, a dropout layer for regularisation, and a fully connected output layer with a sigmoid activation for binary classification. The model was compiled with the binary cross-entropy loss function and the Adam optimiser, and was trained for several epochs with batch processing on a GPU. Validation accuracy was monitored to detect overfitting and to guide the choice of the number of epochs.

4.6 Evaluation Workflow

After training, every model was evaluated on the same test set. Predictions were converted into class labels and compared against the ground truth to produce a confusion matrix, from which accuracy, precision, recall, and F1-score were derived. In addition to predictive performance, training time and approximate memory usage were recorded for each model to support the computational-efficiency comparison reported in the next chapter.

5. MAJOR FINDINGS, OUTCOMES, AND RESULTS

The performance of machine learning and deep learning models for sentiment analysis was evaluated on the IMDb dataset using the metrics described in Chapter 2. This chapter presents and discusses the results obtained, both from a predictive performance perspective and from a computational efficiency perspective.

5.1 Performance Comparison

Table 2 summarises the performance of the four models across accuracy, precision, recall, and F1-score.

Table 2. Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score
Naive Bayes	0.82	0.80	0.84	0.82
Logistic Regression	0.85	0.84	0.86	0.85
SVM	0.86	0.85	0.87	0.86
LSTM	0.90	0.89	0.91	0.90

5.2 Analysis of Machine Learning Models

Among the traditional machine learning models, Support Vector Machine achieved the highest accuracy of 86 percent, followed closely by Logistic Regression at 85 percent and Naive Bayes at 82 percent. Naive Bayes, while computationally efficient, showed slightly lower performance due to its simplifying assumptions about feature independence. Logistic Regression demonstrated balanced performance across all evaluation metrics, making it a reliable baseline model. SVM outperformed the other ML models due to its ability to handle high-dimensional feature spaces effectively, especially when combined with TF-IDF features.

5.3 Analysis of the Deep Learning Model

The LSTM model achieved the highest overall accuracy of 90 percent, outperforming all machine learning models. This improvement can be attributed to its ability to capture sequential dependencies and contextual information in text data. The training and validation accuracy show

that the model gradually improves over epochs, indicating effective learning. However, the training time for LSTM was significantly higher compared to ML models, and it required substantially more computational resources, including GPU acceleration to be practical.

5.4 Computational Efficiency

Table 3 presents a comparison of training time and resource usage across all models. Machine learning models were significantly faster to train and required less memory, making them suitable for real-time or resource-constrained applications. In contrast, the LSTM model required more time and computational power but delivered superior predictive performance.

Table 3. Computational Efficiency Comparison

Model	Training Time	Resource Usage
Naive Bayes	Very Low	Low
Logistic Regression	Low	Low
SVM	Medium	Medium
LSTM	High	High

5.5 Key Findings

- There is a clear trade-off between accuracy and computational efficiency.
- For large datasets, deep learning models are more effective.
- For smaller datasets or limited resources, machine learning models are more practical.
- Feature engineering plays a crucial role in ML models, whereas DL models learn features automatically from raw text.
- Deep learning models, particularly LSTM, provide higher accuracy and better contextual understanding.
- SVM is the best-performing ML model in this study, while LSTM is the best overall model in terms of predictive performance.

6. CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

This study presented a comparative analysis of machine learning and deep learning models for sentiment analysis using a standard textual dataset. The objective was to evaluate and compare these approaches based on performance metrics, computational efficiency, and suitability for different application scenarios.

The experimental results demonstrate that deep learning models, particularly Long Short-Term Memory networks, outperform traditional machine learning models in terms of accuracy, precision, recall, and F1-score. This superior performance is primarily due to the ability of deep learning models to automatically learn complex patterns, contextual relationships, and sequential dependencies within text data.

In contrast, machine learning models such as Naive Bayes, Logistic Regression, and Support Vector Machines rely heavily on manual feature extraction techniques like TF-IDF, which limits their ability to capture deeper semantic meaning. Among the machine learning models, SVM achieved the best performance, followed by Logistic Regression and Naive Bayes.

A key finding of this study is the trade-off between performance and computational efficiency. While deep learning models provide higher accuracy and better generalisation, they demand greater computational resources and longer training times. In contrast, machine learning models offer faster execution and lower resource consumption, making them more practical for real-time systems and environments with hardware constraints.

In conclusion, both machine learning and deep learning approaches have their own strengths and limitations. The findings of this research emphasise the importance of choosing the right model based on the specific context of the problem. The choice between approaches should be informed not only by raw accuracy but also by deployment constraints, dataset size, and interpretability requirements.

6.2 Future Scope

Although this study provides a comprehensive comparison between machine learning and deep learning models for sentiment analysis, there are several directions in which this research can be further extended and improved.

One important area for future work is the exploration of advanced deep learning architectures, particularly transformer-based models such as BERT and GPT. These models have demonstrated state-of-the-art performance in various NLP tasks due to their ability to capture contextual relationships more effectively than traditional models like LSTM and CNN.

Another potential direction is the development of hybrid and ensemble approaches that combine the strengths of both machine learning and deep learning techniques. Future research can also focus on aspect-based sentiment analysis, which involves identifying sentiments toward specific aspects or features within a text rather than determining overall sentiment.

Multilingual and cross-lingual sentiment analysis presents another promising avenue for further investigation. Most existing models are primarily trained on English datasets, which limits their applicability in diverse linguistic contexts. Extending sentiment analysis models to support multiple languages, especially low-resource languages, would significantly increase their real-world usability.

Furthermore, ongoing research should investigate model optimisation techniques such as pruning, quantisation, and knowledge distillation to reduce model size and improve inference speed without significantly compromising performance, particularly for deploying sentiment analysis in real-time applications and resource-constrained environments.

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APPENDICES

Appendix A: List of Abbreviations

- NLP — Natural Language Processing
- ML — Machine Learning
- DL — Deep Learning
- SVM — Support Vector Machine
- NB — Naive Bayes
- LR — Logistic Regression
- CNN — Convolutional Neural Network
- RNN — Recurrent Neural Network
- LSTM — Long Short-Term Memory
- BERT — Bidirectional Encoder Representations from Transformers
- GPT — Generative Pre-trained Transformer
- BoW — Bag of Words
- TF-IDF — Term Frequency – Inverse Document Frequency
- TP, TN, FP, FN — True Positive, True Negative, False Positive, False Negative

Appendix B: Dataset Information

Dataset name: IMDb movie reviews.

Type: Binary sentiment classification (positive vs. negative).

Split: 70 percent training, 30 percent testing.

Source: standard, publicly available benchmark used widely in sentiment analysis research.

Appendix C: Hyperparameter Notes

Hyperparameters for each model were selected based on a combination of recommended defaults and limited tuning. Logistic Regression used L2 regularisation with the inverse regularisation strength tuned via cross-validation. SVM used a linear kernel, which is well known to be effective

for high-dimensional sparse text features. The LSTM was trained with the Adam optimiser, binary cross-entropy loss, and a dropout layer to mitigate overfitting; the number of epochs was selected by monitoring validation accuracy.